Imaging, shape, and volumetric workflows

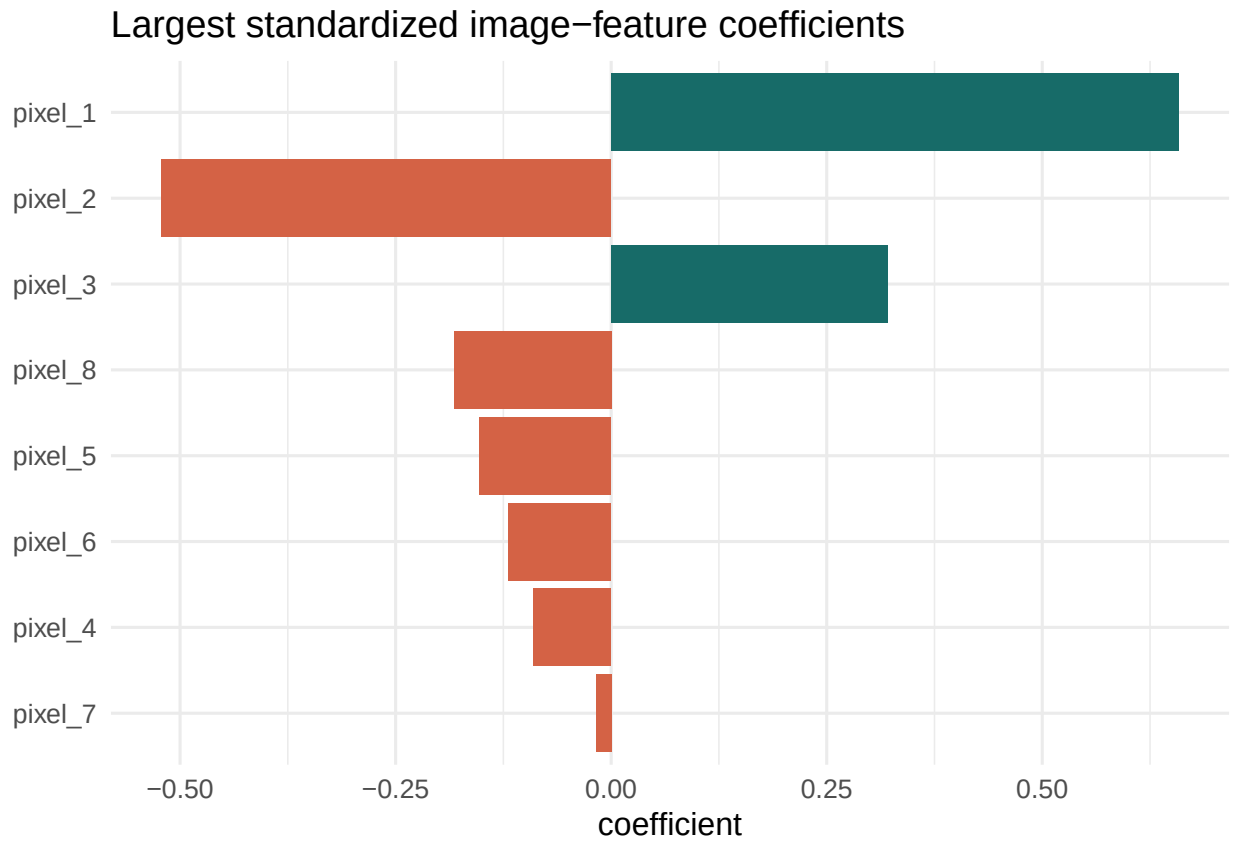
This vignette uses three compact deterministic datasets: 48 image subjects, six contours, and 72 volumetric observations.

Wide-image scalar regression screen

```
image <- demo_image_data(n = 48, features = 8, seed = 2026)
pixel_features <- grep("^pixel_", names(image), value = TRUE)
```

```
scalar <- run_analysis(
  "scalar_image", image, list(outcome = "outcome"), pixel_features,
  list(image_ridge = 5, seed = 2026)
)
scalar$summary
#>       subjects image_features          ridge   heldout_rmse    heldout_mae
#>     48.0000000      8.0000000      5.0000000      0.6494585      0.5956614
head(scalar$table)
#>   feature coefficient
#> 1 pixel_1   0.6577132
#> 2 pixel_2  -0.5211847
#> 3 pixel_3   0.3212493
#> 8 pixel_8  -0.1822060
#> 5 pixel_5  -0.1525856
#> 6 pixel_6  -0.1187539
```

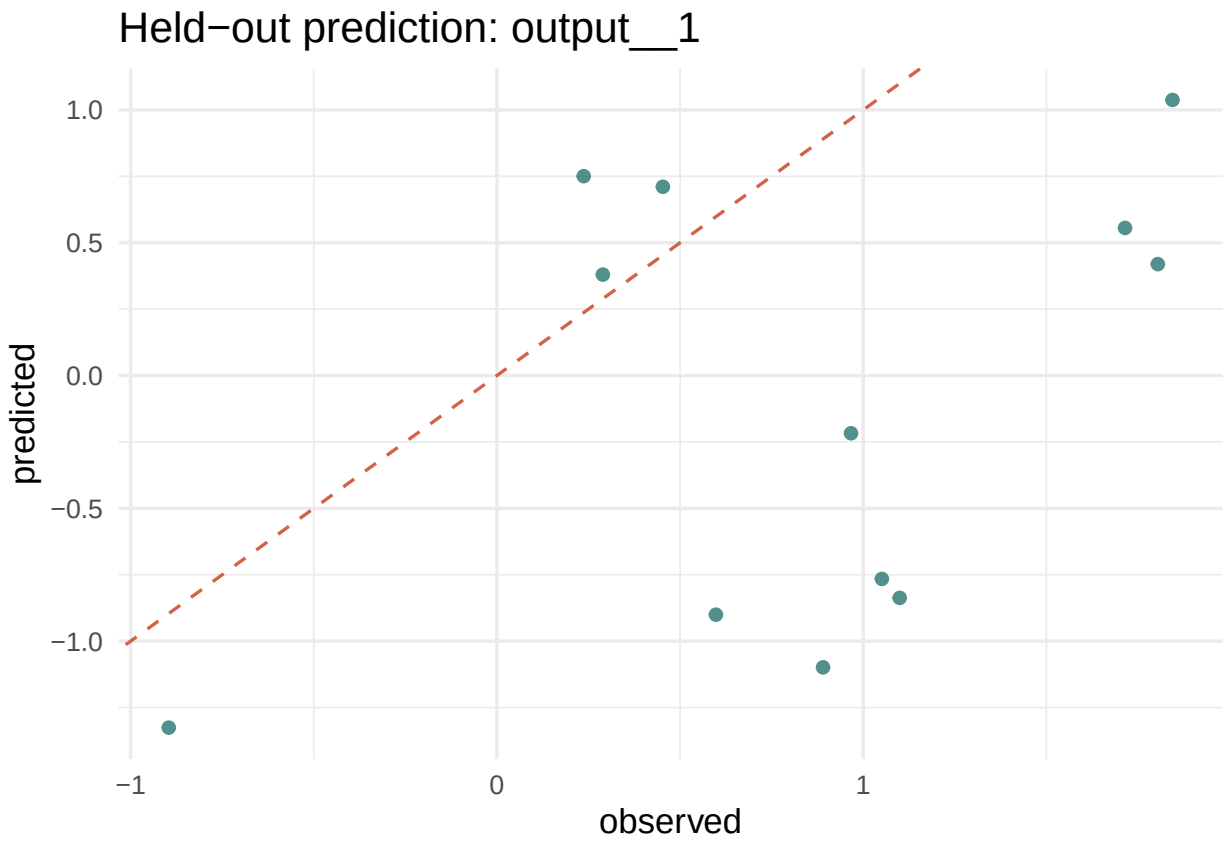
```
scalar$plot
```



Latent image-to-image regression screen

```
paired_features <- grep("^(input__|output__)", names(image), value = TRUE)
latent <- run_analysis(
  "image_to_image", image, list(), paired_features,
  list(latent_factors = 3L, seed = 2026)
)
latent$summary
#>             subjects       input_features      output_features
#>           48.0000000            8.0000000           8.0000000
#>       latent_factors overall_heldout_rmse
#>            3.0000000            0.9418131
head(latent$table)
#>           output_feature heldout_rmse
#> output__1      output__1    1.2612515
#> output__2      output__2    0.7186775
#> output__3      output__3    1.2653351
#> output__4      output__4    0.6910500
#> output__5      output__5    0.5696722
#> output__6      output__6    1.0008603
```

```
latent$plot
```



These fast screens expect already aligned and consistently ordered image features. They are not the full SV-NN, ST-CAR, or SBLF posterior engines.

```
method_citations("scalar_image")
#> [1] "Bhadury, S. (2026). SpatialMethodsWorkbench: Spatial Methods Workbench (Version 0.4.0) [Compute
#> [2] "Hoerl, A. E. and Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal P
#> [3] "Wu, B., Wu, K., and Kang, J. (2025). Bayesian Scalar-on-Image Regression with a Spatially Varyi
#> [4] "Xu, Y. and Kang, J. (2025). Bayesian Image Regression with Soft-thresholded Conditional Autoreg
method_citations("image_to_image")
#> [1] "Bhadury, S. (2026). SpatialMethodsWorkbench: Spatial Methods Workbench (Version 0.4.0) [Compute
#> [2] "Guo, C., Kang, J., and Johnson, T. D. (2022). A Spatial Bayesian Latent Factor Model for Image-
```

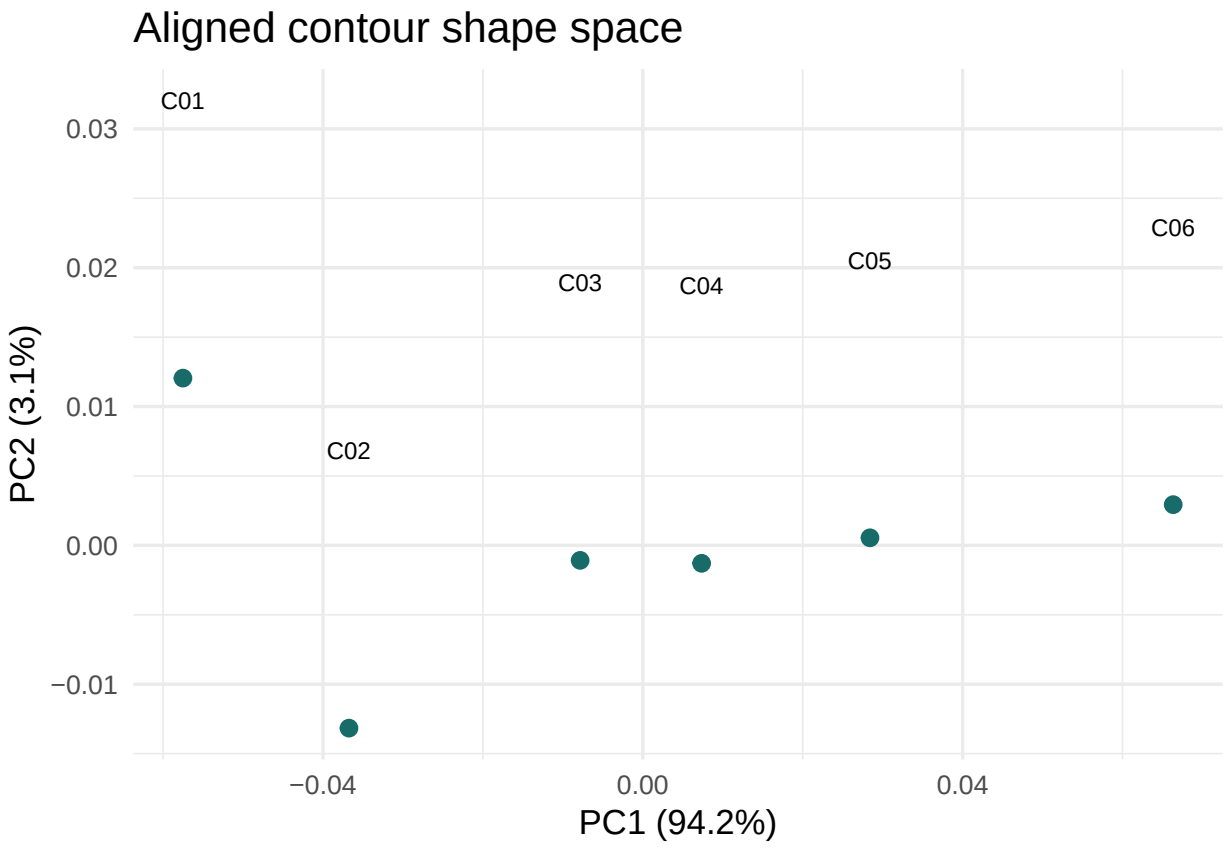
Ordered-contour shape PCA

```
contours <- demo_contour_data(contours = 6, points = 16, seed = 2026)
head(contours)
#>   contour_id order             x         y
#> 1        C01     1  1.023845e+00 0.4031477
#> 2        C01     2  7.602709e-01 0.7227267
#> 3        C01     3  3.686349e-01 0.8460146
#> 4        C01     4  5.897198e-17 0.9155559
#> 5        C01     5 -3.841713e-01 0.8816706
#> 6        C01     6 -7.758343e-01 0.7375215
```

```
shape <- run_analysis(
  "shape_pca", contours,
```

```
  list(id = "contour_id", x = "x", y = "y"), character(),
  list(landmarks = 16L)
)
shape$summary
#>     contours    landmarks pc1_variance pc2_variance
#>     6.000000    16.000000    94.150808     3.113624
shape$table
#>       id          PC1           PC2
#> C01 C01 -0.057519448  0.01204777771
#> C02 C02 -0.036752447 -0.0131625653
#> C03 C03 -0.007838020 -0.0010771355
#> C04 C04  0.007357972 -0.0012901654
#> C05 C05  0.028412553  0.0005464276
#> C06 C06  0.066339390  0.0029356615
```

```
shape$plot
```



Input points must follow each boundary in order. The vignette reports reproducible scores without assigning a tumor phenotype.

Volumetric ISPAT-3D input

```
volume <- demo_spatial_3d_data(n = 72, volumes = 2, seed = 2026)
volume_features <- c("epithelial", "APC", "CTL", "Treg", "T_helper")
volume_mapping <- list(
  x = "x", y = "y", z = "z", zone = "zone", id = "volume_id"
```

```
)
validate_dataset(volume, volume_mapping)[c("ok", "n", "p")]
#> $ok
#> [1] TRUE
#>
#> $n
#> [1] 72
#>
#> $p
#> [1] 10
```

The public ISPAT3D engine (CRAN, 0.3.1 or newer) is opt-in and is not executed during vignette builds. The path calls `ISPAT3D::ispat3d_fit()`: an anisotropic Matérn-3/2 Vecchia GP per marker pooled within each zone over registered x, y, z, then the shared-plus-zone covariance likelihood. The two `volume_id` values are used as section labels for anchor balancing. The settings below are smoke values for 72 rows; omit them to use the package defaults.

```
install.packages("ISPAT3D")

ispat3d <- run_analysis(
  "ispat3d", volume, volume_mapping, volume_features,
  list(
    rank = 2L, anchor_min = 12L, anchor_max = 24L, gp_neighbors = 5L,
    gp_maxit = 3L, factor_maxit = 60L, threads = 1L, seed = 2026L,
    threshold = 0.05
  )
)
ispat3d$summary
names(ispat3d$matrices)    # Low, Intermediate, High
head(ispat3d$table)
table(ispat3d$gp_log$status)
```

The 72-row dataset is a software demonstration. Registered coordinates, section alignment, zones, and study-specific sampling must be established before using real volumetric data.